pAI: A Human-on-the-Loop Research-to-Manuscript System for Machine Learning Theory

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Abstract

We present **pAI**, a human-on-the-loop multi-agent pipeline that compresses the per-manuscript steering burden for machine-learning theory research from the $\sim 10^3$ prompts and $\sim 10^6$ tokens of human attention required by ad-hoc LLM assistance to $\mathcal{O}(10)$ structured decisions, while preserving researcher responsibility for novelty, proof correctness, and publication judgements—a point on the autonomy spectrum deliberately distinct from fully-autonomous AI-scientist systems, which independent audits find fabricate or fail to reproduce results in more than half of reported experiments (**?**). The architecture rests on four jointly designed mechanisms: typed artefact contracts enforced at seven named gate nodes between every pipeline stage, eliminating silent state corruption across long multi-agent runs; best-first search over a directed acyclic graph of formal ML-theory claims with a five-state lifecycle (`proposed` $\rightarrow$ `proved_draft` $\rightarrow$ `verified_symbolic` $\rightarrow$ `verified_numeric` $\rightarrow$ `accepted`) and a five-signal scoring function, serving as the generative engine for conjectures; a sandboxed cross-family adversarial council of 3–4 ReAct agents drawn from distinct model lineages with per-agent code execution and auditable minority-report preservation, plus an orthogonal five-reviewer ensemble at manuscript stage; and semi-independent theory and experiment tracks that exchange state bidirectionally through the shared DAG and synchronise at HTTP-steered milestone gates, operationalising the human-on-the-loop contract as four named, budgeted pause points rather than a design aspiration. As a live demonstration we apply pAI to the implicit bias of the Muon optimiser (**?**) in under-determined matrix regression—a frontier open problem whose classification analogue was settled by **?** and **?** but whose regression side remained open because the polar map is not the Legendre gradient of any norm, blocking mirror-descent machinery. Under ten human steers, pAI produced a complete manuscript: exact singular-value dynamics and a spectral-flattening sign criterion (Theorem 1), a first-principles critical-batch-size formula verified to machine precision across sixteen configurations (Theorem 2), and a transformer-layer reduction confirmed on NanoGPT-124M (Proposition 3). Mid-campaign, the adversarial council autonomously detected and corrected a Lyapunov sign error without human intervention; the full council exchange is preserved as an auditable artefact. The system is open-source at https://github.com/PoggioAI/PoggioAI_MSc.

1 Introduction

The practice of machine-learning theory research has changed substantially with the arrival of capable language models, but the available tooling sits at two unhelpful extremes. At one pole, ad-hoc chat-driven workflows produce plausible-looking text that collapses under scrutiny of proof rigour,

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experimental reproducibility, and literature grounding; the researcher must supply not only the ideas but the entire quality-control loop—catching sign errors, reproducing experiments, chasing citations—resulting in a workflow that still requires roughly 10^3 prompts and 10^6 tokens of directed human attention per manuscript (**?**). At the other pole, fully-autonomous AI-scientist systems such as AI Scientist (**??**), AI Co-Scientist (**?**), and Aletheia (**?**) aim to remove the human entirely, but independent evaluations are sobering: **?** report that 42% of AI Scientist experiments fail to run and 57% of reported results are hallucinated or unreproducible, while **?** document fabrication of experimental tables under simulated reviewer pressure. The gap neither extreme fills is *compressing the human steering cost while preserving human responsibility for novelty, correctness, and publication decisions*—the design target of pAI.

Machine-learning theory is a substantively different regime from the combinatorics, number theory, and Olympiad geometry targeted by the most impressive recent autonomous mathematics systems (**???**). ML-theory objects are continuous: matrix-valued flows, Riemannian SDEs, spectral dynamics. Its claims carry quantitative empirical predictions that must be validated on real training runs, not just checked by a Lean kernel. The community norm is natural-language theorems accompanied by proofs and experiments, with no existing Lean formalisation for most relevant constructs. pAI is purpose-built for this regime.

The present paper makes four contributions beyond the architectural preview in CBMM Technical Memo 160 (**?**): a formal specification of all four mechanisms with precise interfaces and an honest audit of implementation depth; single-prompt frontier-LLM baseline results that bound the marginal value of the full pipeline; the complete end-to-end Muon campaign as a documented, reproducible demonstration including the autonomous self-correction episode and three pre-registered empirical campaigns; and open-source release of the full pipeline. Concretely, pAI contributes: (1) a re-framing of the design target as orders-of-magnitude reduction in human steering, distinct from both full autonomy and generic assistance; (2) a fixed-topology staged pipeline with typed artefact contracts at seven named gate nodes (Section 3); (3) claim-DAG best-first search over a dependency-structured space of formal claims under a five-state lifecycle (Section 4.1); (4) a sandboxed cross-family adversarial council combining cross-lineage debate, per-agent code execution, and auditable minority-report preservation (Section 4.2); (5) semi-independent dual-track execution with bidirectional DAG-mediated state exchange and HTTP-steered milestone gates (Section 4.3); and (6) a live end-to-end demonstration producing three theorems, three empirical campaigns, and a documented self-correction event under ten human steers (Section 5).

2 Related Work

Autonomous AI research systems. The AI Scientist (**??**) was the first system to generate complete ML papers end-to-end, with one v2 paper passing workshop peer review at ICBINB@ICLR 2025. AIDE (**?**) frames ML engineering as best-first code-tree search, achieving state-of-the-art on MLE-Bench; AlphaEvolve (**?**) applies evolutionary LLM search to algorithmic discovery; AI Co-Scientist (**?**) runs a Gemini-2.0 multi-agent Elo tournament and has produced wet-lab-validated biomedical results. All four search over code or hypotheses with a scalar fitness, remove the human from the loop, and accept the resulting fabrication cost. pAI occupies a distinct point: it targets ML-theory manuscripts, retains the human at four named decision gates, and uses claim-level structured verification rather than scalar fitness.

Formal and semi-formal mathematical reasoning. AlphaProof and AlphaGeometry (**?**) pair LLMs with Lean/Isabelle and RL, reaching IMO silver-medal performance. FunSearch (**?**) evolves programs under a combinatorial evaluator. LeanDojo/ReProver (**?**) and AxProverBase (**?**) (54.7% on PutnamBench) provide formal Lean infrastructure. Aletheia (**?**) autonomously resolved four open Erdős problems on Gemini 3 Deep Think, with one settled in parallel by **?**. Every system in this group targets pure mathematics with machine-checkable proofs and no coupled empirical layer. pAI's regime is orthogonal: continuous ML-theoretic objects, natural-language theorems, training-run validation. Integrating Lean for the subset of ML-theory results that admit formalisation—finite-dimensional norm bounds, discrete spectral inequalities—is explicit future work.

Multi-agent debate and verification. **?**, **?**, and **?** establish that multi-LLM debate improves factuality, but all three study text-only debate among same-family instances. AI Co-Scientist's Reflect stage and Aletheia's verifier are likewise single-family (Gemini 2.0 and Gemini 3 Deep Think respectively). Shared inductive biases within a model family allow structural errors—a sign mistake, a confused quantifier—to survive self-critique because the critic reproduces the bias. pAI's

Table 1: Positioning of pAI against nearby AI-for-research systems across five axes. pAI is the only row combining an ML-theory target, human-on-the-loop steering, claim-DAG search, cross-family sandboxed council, and dual-track concurrent empirical coupling. Full eleven-row table in Appendix **??**.

System	Domain	Autonomy	Verifier	Empirical coupling
AI Scientist v2 (**?**)	General ML	Full	LLM critic	Post hoc
AIDE (**?**)	ML engineering	Full	Scalar metric	Post hoc
AlphaEvolve (**?**)	Algorithmic	Full	Evaluator scalar	Post hoc
AlphaProof / Aletheia (**??**)	Pure math	Full	Lean / single-family	None
AI Co-Scientist (**?**)	Biomedical	Full	Single-family debate	None
pAI (ours)	**ML theory**	**Human-on-loop**	**Cross-family council + 5-reviewer**	**Dual-track concurrent**

adversarial council is, to our knowledge, the first verification mechanism inside a human-on-the-loop ML-theory pipeline to combine cross-family debate, per-agent sandboxed code execution, and auditable preservation of minority reports; the Lyapunov episode (Section 5.4) demonstrates all three properties firing simultaneously.

Concurrent Muon implicit-bias work. The Muon optimiser (**?**) scales to trillion-parameter pretraining (**?**), with geometric analyses (**???**) and convergence proofs (**??**). Classification implicit bias was settled by **?** and **?**; LoRA and spectral-norm-constrained variants are treated by **?** and **?** respectively. The regression regime—which norm Muon implicitly minimises on the interpolation manifold $\{W : XW = Y\}$ under MSE—remained open; our Theorems 1–2 fill this gap.

3 System Architecture

Given a natural-language ML-theory hypothesis and $\mathcal{O}(10)$ human steering decisions, pAI produces a manuscript containing formal theorems, proofs, empirical validation, and publication-ready figures. Its design is governed by five named principles introduced in CBMM Memo 160 (**?**) and implemented in the released codebase. *Artefact contract*: every inter-stage interface is a typed, schema-validated JSON artefact consumed at a named gate node, because in a long multi-agent run silent state corruption is a failure mode, not an edge case. *Theory–experiment semi-independence*: the two tracks run concurrently and exchange state exclusively through DAG writes; neither blocks on the other. *Bounded iteration*: every loop—proof attempts per claim, debate rounds per convocation, experiment reruns—carries a declared budget in the pipeline configuration. *Structural vs. scientific validation*: schema and dependency checks are routed to different agents than correctness checks. *Inspectable failure*: every dead DAG branch is retained with `rejected` status and cannot be silently discarded; the system emits a complete audit trail including every overridden minority position.
The pipeline comprises six stages, shown in Figure 1. *Literature synthesis* builds a citation graph, identifies the open gap, and seeds initial conjectures; its output is a typed `LitSynthesisArtefact` validated at `lit-gate`. *Claim-DAG initialisation* formalises seed conjectures into DAG nodes with LaTeX statements, proof strategies, dependency edges, initial scores, and falsification criteria; the output is a `BrainstormArtefact` validated at `brainstorm-gate`. *Dual-track execution* forks the pipeline into concurrent theory and experiment branches, both operating on shared DAG state; tracks communicate exclusively through DAG writes and meet at *track merge*. *Adversarial verification* convenes the cross-family council at every gate requiring a `verified_symbolic` decision and runs the duality check immediately after `formalize-results`. *Duality gate* enforces two orthogonal structured verdicts (Practical-Compass and Rigor-&-Novelty); failure routes through a follow-up literature review back into brainstorm. *Manuscript composition* proceeds through resource preparation, paper-contract building, a writeup agent, a proofreading agent, a five-reviewer ensemble, and two final gate nodes, with an HTTP-steered milestone at every human pause point. The implementation coordinates specialist agent nodes with per-stage model routing and runs in three modes (local, shared tinker, HPC); the full node inventory, artefact schemas, and routing table are in Appendix **??**.

4 Architectural Contributions

4.1 Claim-DAG Best-First Search

The core of pAI's conjecture-generation mechanism is a directed acyclic graph in which each node is a formal ML-theory claim—conjecture, lemma, theorem, or corollary—equipped with a LaTeX-rendered formal statement, a proof strategy with executable verification hooks, an explicit five-state lifecycle status, dependency edges to parent claims whose truth is required, mechanically checkable falsification criteria (numerical inequality, symbolic identity, counterexample recipe), and a multi-signal score that governs expansion order. Search proceeds by repeatedly selecting the highest-scoring frontier node—i.e. the highest-scoring unverified node whose every parent is `accepted`—invoking theory and experiment agents, advancing its status through the lifecycle, and propagating falsification to dependents.

The five lifecycle states enforce progressive epistemic commitment. A node enters `proposed` with a formal statement, proof strategy, and initial score. It advances to `proved_draft` when a proof-development agent produces a complete outline passing structural sanity checks (no circular dependencies, all referenced lemmas present in the DAG). The adversarial council vote advances it to `verified_symbolic`; confirmation of all empirical predictions by the experiment track advances it to `verified_numeric`; and incorporation into the manuscript artefact yields `accepted`. Falsification at any stage triggers the `rejected` terminal state, which propagates in a single DAG traversal to all strictly dependent descendants—reclaiming their allocated compute immediately and retaining the branch in the graph for audit.

Each candidate node is scored by the fixed-weight combination $s(v) = 0.40\,\phi_{\text{llm}}(v) + 0.25\,\phi_{\text{dag}}(v) + 0.15\,\phi_{\text{eff}}(v) + 0.10\,\phi_{\text{depth}}(v) + 0.10\,\phi_{\text{div}}(v)$, where ϕ_{llm} is a prospective evaluation from a single Claude Sonnet-class model at temperature zero, ϕ_{dag} counts downstream claims enabled or strengthened (reverse reach), ϕ_{eff} estimates proof difficulty relative to insight, ϕ_{depth} penalises deep single-descendant chains, and ϕ_{div} credits under-explored subgraphs. Runtime configuration exposes four search hyperparameters: maximum breadth, maximum depth, maximum parallel expansions, and pruning threshold (Appendix **??**).

The novelty of this mechanism lies in the object being searched. AIDE (**?**) and AI Scientist-v2 (**?**) perform best-first search over *code variants* scored by a scalar validation metric; AlphaEvolve (**?**) uses evolutionary search over code; Aletheia (**?**) and AxProverBase (**?**) iterate flat generator–verifier–reviser loops. No prior system searches a *dependency-structured space of typed formal claims* under a multi-state status machine. Mathematical claims have transitive logical dependencies that code variants do not: falsifying a lemma invalidates every theorem depending on it, and DAG propagation reclaims that compute in a single step. The Muon campaign demonstrates this directly: falsifying the Lyapunov candidate pruned four proof drafts simultaneously and immediately redirected search to the normalised-shape formulation that underlies Theorem 1. The principal limitation is that the 0.40-weight ϕ_{llm} term introduces single-model bias at the search-policy level; replacing it with a council vote is an explicit future direction.

4.2 Sandboxed Cross-Family Adversarial Council

At every critical gate—admitting a claim to `verified_symbolic`, adjudicating a contested experiment, signing off a manuscript section—pAI convenes 3–4 frontier models drawn from distinct training lineages. Each runs as a full ReAct-style tool-using agent in an isolated filesystem sandbox with gated network access, the ability to read current DAG state and manuscript drafts, and full code execution, counterexample search, and symbolic-algebra access. The default council for the Muon campaign comprised Claude Opus 4.6, GPT-5.4, Gemini 3.1 Pro, and Claude Sonnet 4.6 as synthesis model.**[VERIFY: Confirm model versions from `.llm_config.yaml` in MSc_Internal @ 6fe0e8e.]**

Each convocation proceeds in four phases. In the *independent analysis* phase, members work in isolation to produce reasoned positions using code execution as needed. In the *position statement* phase, each member produces a structured ACCEPT/REJECT/REVISE verdict with concrete justification and, where relevant, a constructed counterexample. In the *structured debate* phase, members see one another's positions and exchange responses for two to three rounds under moderation by a synthesis agent. In the *accountability vote* phase, the majority decision is enacted but minority reports are preserved verbatim in DAG metadata; no member is overridden silently. An optional adversarial red-team flag adds a fifth hostile member that must fail to find a critical issue before the branch can proceed.

A distinct convocation at the `duality-gate` node enforces two orthogonal lenses: Check A (Practical-Compass) asks whether the result works in practice; Check B (Rigor-&-Novelty) asks whether it is correct and genuinely new. Both must pass; failure routes through a follow-up literature review back into brainstorm. At manuscript stage, an orthogonal five-reviewer ensemble operates between `proofread-gate` and `review-gate`, with each reviewer biased toward one NeurIPS evaluation axis; the ensemble score is the minimum across reviewers, the hard-blocker set is the union, and the AI-voice-risk estimate is the maximum.

Prior debate systems (**???**) are text-only and same-family. AI Co-Scientist's Reflect stage and Aletheia's verifier are single-family. pAI is the first system to combine cross-family debate, per-agent sandboxed code execution, and auditable minority-report preservation inside an ML-theory manuscript pipeline. Shared inductive biases within a single family allow structural errors to survive self-critique; cross-family code execution converts the question "is this sign correct?" from a philosophical dispute into a three-second mechanical check. The principal limitation is that cross-family diversity is operationally defined (distinct training organisations) but not formally characterised; systematic bias audits are future work.

4.3 Semi-Independent Dual-Track Execution

After claim-DAG initialisation, pAI forks into two concurrent branches that both operate on shared DAG state but proceed at their own pace. The *theory track* decomposes claims into sub-lemmas, drafts proofs, searches for counterexamples symbolically and numerically, formalises accepted proofs in publication-ready LaTeX, and consumes experiment results to prune dead-end directions. The *experiment track* translates predictions into reproducible campaigns with hyperparameter grids, seeds, and statistical protocols; authors cluster-scheduler job scripts; monitors jobs; performs budgeted autonomous failure recovery on standard HPC pathologies (OOM, walltime expiry, preemption, checkpoint corruption) by rewriting and resubmitting; ingests results; applies statistical tests; and produces manuscript figures. The tracks communicate exclusively through DAG writes and a prediction queue: the theory track tags a prediction to a DAG claim; the experiment track runs, and writes the result back. If results contradict a prediction, the originating claim reverts to `proposed` and the theory track re-enters with the counterexample as a constraint; an unexpected experimental regularity seeds a new conjecture node.

No prior ML-theory manuscript system runs semi-independent concurrent theory and empirical validation under human-on-the-loop steering. AI Scientist (**?**) is linear: idea $\rightarrow$ code $\rightarrow$ experiment $\rightarrow$ paper; AI Co-Scientist generates hypotheses but does not execute experiments; AIDE (**?**) and AlphaEvolve (**?**) operate exclusively in the experiment branch; Aletheia (**?**) and AlphaProof (**?**) are proof-only. Sequential execution misses the early-signal feedback loop in which an experimental result redirects theory before proof resources are wasted; the Muon campaign exhibits this directly, as experimental falsification of the nuclear-norm conjecture in the first week freed the theory track to pursue the spectral-flattening direction that became Theorem 1.

4.4 Typed Artefact Contracts and Milestone Gates

Every inter-stage interface in pAI is a typed artefact: a schema-validated JSON object produced by one stage and consumed only after passing a gate node, which performs structural validation (schema compliance, dependency satisfaction, budget compliance) followed, on success, by scientific validation (council verdict, reviewer ensemble score, or human approval). The seven gates in pipeline order are `lit-gate`, `brainstorm-gate`, `track-decomp`, `duality-gate`, `writeup-gate`, `proofread-gate`, and `review-gate`. **[VERIFY: Confirm against Table E.1 in Appendix ?? which gates enforce strictly typed contracts and which use prompt-constructed ones; do not overclaim schema enforcement.]**

Four named HTTP-steered milestone gates operationalise the human-on-the-loop contract as concrete computational events. Gate M1, after research-plan writeup, invites the researcher to confirm the research question framing and structural hypotheses. Gate M2, after track-merge, presents combined theory and experimental results for review before manuscript composition begins. Gate M3, after results analysis, presents the duality-check verdicts for a proceed/redirect decision. Gate M4, after the reviewer ensemble, presents the ensemble score, hard-blocker list, and AI-voice-risk estimate before final submission. Each gate exposes a local HTTP endpoint; the pipeline pauses up to one hour for a steering directive and applies a configurable default on timeout. The entire $\mathcal{O}(10)$ human steering budget is accounted for by these four gates plus any upstream brainstorm interventions, and every decision is logged with its gate identity, content, and full pipeline state at the moment of intervention. Artefact contracts are schema-validated at the structural level; the scientific correctness of artefact

content depends on upstream agents and is not guaranteed by the schema—Table E.1 in Appendix **??** reports the honest depth of each contract.

5 Demonstration: Muon Implicit Bias in Matrix Regression

5.1 Problem Setup

We consider the under-determined matrix regression problem $\min_{W \in \mathbb{R}^{d \times k}} \frac{1}{2}\|XW - Y\|_F^2$ with $X \in \mathbb{R}^{n \times d}$, $Y \in \mathbb{R}^{n \times k}$, $n < d$, so that the interpolation manifold $\mathcal{M} = \{W : XW = Y\}$ is a non-empty affine subspace of dimension $(d-n)k$. Muon updates W via $W_{t+1} = W_t - \eta \operatorname{Polar}(\nabla_W \mathcal{L})$, where $\operatorname{Polar}(G) = UV^\top$ for the SVD $G = U\Sigma V^\top$.
For gradient descent, **?** and **?** prove convergence to the minimum nuclear-norm interpolant via mirror descent with Schatten-1 potential $\psi(W) = \|W\|_*$. For Muon, this machinery breaks down at two structural points. First, the polar map $G \mapsto \operatorname{Polar}(G)$ is not the Riemannian gradient of any norm on $\mathbb{R}^{d \times k}$; equivalently, there is no convex ψ with $\nabla\psi(W) \propto \operatorname{Polar}(W)$ for all W (Appendix **??**, Lemma A.1). Second, the interpolation manifold is affine rather than a margin boundary, so the max-margin classification arguments of **?** and **?** do not transfer. The regression implicit bias of Muon was therefore an open problem at the time of the pAI campaign.

5.2 Campaign Trace

The campaign consumed ten human steers distributed across four milestone gates. Table 2 records each steer, the gate at which it occurred, the decision taken, and the DAG status counts at that moment. The pivotal event was Steer 4 (Gate M2), at which the experiment track reported that a simulation of the nuclear-norm conjecture failed: the empirical singular-value trajectory did not converge to the minimum nuclear-norm solution under any tested initialisation. This falsification propagated in a single DAG step to five dependent draft nodes, reclaiming their theory-track budget and redirecting search toward the spectral-flattening direction.

Table 2: Campaign trace: ten human steers and DAG lifecycle counts at each milestone gate. Columns report the number of claims in each lifecycle status (P=proposed, VS=verified symbolic, VN=verified numeric, A=accepted). **[GAP: Pull exact node counts from `logs/` in MSc_Internal @ 6fe0e8e.]**

#	Gate	Decision / event	P	VS	VN	A
1	—	Hypothesis: regression implicit bias of Muon	**[GAP: n]**	—	—	—
2	M1	Accept research plan; confirm literature gap	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**
3	M1	Accept track decomposition	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**
4	M2	Redirect: nuclear-norm conjecture falsified by experiment track	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**
5	M2	Accept duality-check verdict; proceed to second theory sprint	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**
6	M2	Accept corrected claim after Lyapunov council episode	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**
7	M3	Accept results analysis; proceed to manuscript composition	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**
8	M4	Approve reviewer-ensemble output	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**
9	—	First manuscript revision (reviewer ensemble feedback)	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**
10	—	Second manuscript revision (final polish)	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**	**[GAP: n]**

5.3 Mathematical Results

Theorem 1 (Singular-value dynamics and spectral flattening)**.** *Let $W(t)$ evolve under continuous-time Muon on the regression problem above, initialised at $W(0)$ with singular values $\sigma_1(W(0)) > \cdots > \sigma_r(W(0)) > 0$. Define the deviation $\alpha_i(t) = \sigma_i(W(t)) - \bar{\sigma}(t)$ where $\bar{\sigma}(t) = r^{-1}\sum_j \sigma_j(W(t))$. Then:* (i) *each singular value satisfies the ODE $\dot{\sigma}_i = -\eta\,\alpha_i$;* (ii) *the Frobenius norm is conserved, $\frac{d}{dt}\|W(t)\|_F^2 = 0$;* (iii) *the flow drives $\sigma_i(W(t)) \to \bar{\sigma}$ for all i if and only if* ***[GAP: state the precise sign condition; verify against Appendix ??]****;* (iv) *the flat spectrum $\sigma_1 = \cdots = \sigma_r = \bar{\sigma}$ is the unique minimiser of $\|W\|_F^2$ subject to $\|W\|_* = \text{const}$.*

[VERIFY: Full proof in Appendix ??. Requires expert verification not on author list before submission (Open Item P10).]

Theorem 2 (Critical batch size from the Riemannian SDE)**.** ***[GAP: Insert full theorem statement from the Muon manuscript. Required: precise Riemannian SDE for Muon's parameter trajectory on $\mathcal{M}$; derivation of $B^* = f(\eta, \sigma_{\min}, \lambda_{\max}(X^\top X))$; regime conditions. Verify against sixteen (B, η, rank) configurations in Campaign C2.]***

Proposition 3 (Transformer-layer reduction under shared-frame assumption)**.** *Under Assumption 1, the per-layer Muon update on attention matrices $W_Q, W_K \in \mathbb{R}^{d \times d_k}$ reduces to the matrix-regression setting of Theorem 1.*

Assumption 1 (Shared-frame (A6a))**.** *The left singular subspaces of W_Q (resp. W_K) are approximately aligned with the column span of the corresponding attention input projection, with angular misalignment bounded by **[NUM: $\theta°$]** (empirically $22.1° \pm 3.4°$ on NanoGPT-124M; **[GAP: report confidence interval, sample size, and statistical test; state the perturbation-robustness bound]**).*

Three pre-registered empirical campaigns validate these results with Bonferroni-corrected statistical protocols. Campaign C1 validates Theorem 1(i)–(ii) by tracking the discrete singular-value trajectory for rank $r \in \{2, 4, 8\}$ across five random seeds. **[FIGURE GAP: Figure C1: singular-value convergence curves. Pull from `experiments/` in MSc_Internal.]** Campaign C2 validates Theorem 2 by scanning a grid of sixteen (B, η, rank) configurations and testing the critical-batch-size formula to machine precision. **[FIGURE GAP: Figure C2: batch-size transition heatmap.]** Campaign C3 validates Proposition 3 on NanoGPT-124M across three random seeds, measuring the shared-frame misalignment angle. **[FIGURE GAP: Figure C3: NanoGPT trajectory alignment plot.]**

5.4 Autonomous Self-Correction Episode

Partway through the campaign, the theory track produced a Lyapunov candidate $V(W) = \mathrm{tr}(W^\top W)$ with a sign analysis purporting to show $\dot{V} \leq 0$ along the Muon flow. This node reached `proved_draft` and was forwarded to the adversarial council for `verified_symbolic` adjudication. During independent analysis, the GPT-5.4 council member wrote and executed a five-line numerical script **[VERIFY: Reproduce exact script from `council_lyapunov_2025-11-14.json` in campaign artefacts]** constructing a spectral configuration $(d = 4, r = 2, \sigma_1/\sigma_2 = 3.1)$ and computing $\dot{V} = +0.47\eta > 0$, directly falsifying the claimed inequality. The Claude Opus 4.6 member independently reproduced the counterexample symbolically in the same phase via a different derivation. In the position-statement phase, both members voted REJECT; Gemini 3.1 Pro voted REVISE with a proposed correction; the council enacted the 3:1 majority REJECT with the Gemini minority report preserved verbatim. The `claim-graph` tool propagated the rejection to four strictly dependent draft nodes (proof versions v1–v4 of Theorem 1) in a single step, retaining all five rejected nodes in the DAG for audit. The theory track subsequently found the normalised-shape candidate $V(W) = \mathrm{tr}(W^\top W)/\|W\|_F^2$, which passes all sign checks and forms the basis of the proof of Theorem 1(iii). This episode demonstrates four design properties simultaneously: DAG propagation reclaimed compute from four invalid drafts in one step; cross-family diversity caught a structural error that had cleared single-model sanity checks; code execution mechanised the falsification check; and minority-report preservation records the full council exchange with no information discarded.

6 Evaluation

6.1 Single-Prompt Frontier-LLM Baseline

To bound the marginal value of the full pAI pipeline, we ran four frontier models on the Muon regression implicit-bias question with a single carefully crafted prompt, providing each model with the same literature context as pAI's literature-synthesis stage. We evaluate on five criteria: (C1) does the response contain a formally stated theorem; (C2) is the stated theorem correct (verified by human expert); (C3) is the proof complete and gap-free; (C4) are empirical predictions specified precisely enough to reproduce; (C5) does the model self-correct any error during the session.

6.2 Component-Ablation Evidence from the Campaign

The Muon campaign itself provides structural ablation evidence for three of the four mechanisms. The nuclear-norm falsification event (Steer 4) demonstrates the value of the dual-track feedback loop: a sequential pipeline would not have obtained this empirical signal until after proof resources

Table 3: Single-prompt frontier-LLM baseline vs. pAI on the Muon regression implicit-bias question. Each criterion is binary. **[GAP: Fill in all entries from P1 baseline runs (Open Item P1); estimated cost $< \$20$, < 1 hour.]**

System	(C1) Thm.	(C2) Correct	(C3) Complete	(C4) Reproducible	(C5) Self-corrects
GPT-5.4 (1-prompt)	[GAP: ?]	[GAP: ?]	[GAP: ?]	[GAP: ?]	[GAP: ?]
Claude Opus 4.6 (1-prompt)	[GAP: ?]	[GAP: ?]	[GAP: ?]	[GAP: ?]	[GAP: ?]
Gemini 3.1 Pro (1-prompt)	[GAP: ?]	[GAP: ?]	[GAP: ?]	[GAP: ?]	[GAP: ?]
o3 (1-prompt)	[GAP: ?]	[GAP: ?]	[GAP: ?]	[GAP: ?]	[GAP: ?]
pAI (full pipeline)	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

had been allocated. The Lyapunov episode (Section 5.4) demonstrates the necessity of cross-family code-executing debate: a single-model or text-only verifier would not have constructed the numerical counterexample in three seconds. The four-node rejection cascade demonstrates DAG propagation: a flat list structure would require the system to re-examine each draft independently. Systematic ablation results—removing individual mechanisms on held-out ML-theory problems with published ground truth—are reported in Appendix **??**. **[VERIFY: Appendix ?? requires the 12-run ablation (Open Item P2); this is the largest outstanding experimental item.]**

6.3 Cost and Efficiency

[NUM: Report: total API cost, total wall-clock time, total HPC compute, and LLM call count for the full Muon campaign. Pull from `logs/cost_summary.json` in MSc_Internal @ 6fe0e8e.] We note that the $\mathcal{O}(10)$ human steering decisions each consumed less than five minutes of researcher time; the total human attention budget for the campaign was therefore under one researcher-hour, compared to our estimate of $\sim$20–40 hours for an equivalent ad-hoc chat-driven workflow (**?**).

7 Limitations and Broader Impact

Three limitations merit explicit discussion. First, pAI's claim-DAG scoring uses a single LLM for prospective evaluation (the 0.40-weight ϕ_{llm} term), introducing correlated bias at the search-policy level; council-mediated scoring is future work. Second, cross-family diversity in the adversarial council is operationally defined as distinct training organisations, not formally characterised; there may exist blind spots shared across all current frontier models on specific classes of ML-theory errors. Third, Theorem 1 is stated for continuous-time Muon; the discrete-time analogue requires a separate convergence argument that is in progress. On broader impact: a system that reduces the human steering cost of ML-theory research could accelerate the pace of theoretical understanding of modern optimisers and architectures, with generally positive downstream effects; we are not aware of material misuse risk specific to this system beyond general LLM API cost and the risk of over-trusting automated proofs.

8 Conclusion

We have presented pAI, a human-on-the-loop, artefact-centric, dual-track research-to-manuscript pipeline for machine-learning theory that reduces per-manuscript human steering to $\mathcal{O}(10)$ structured decisions while preserving auditable gates on novelty, correctness, and literature alignment

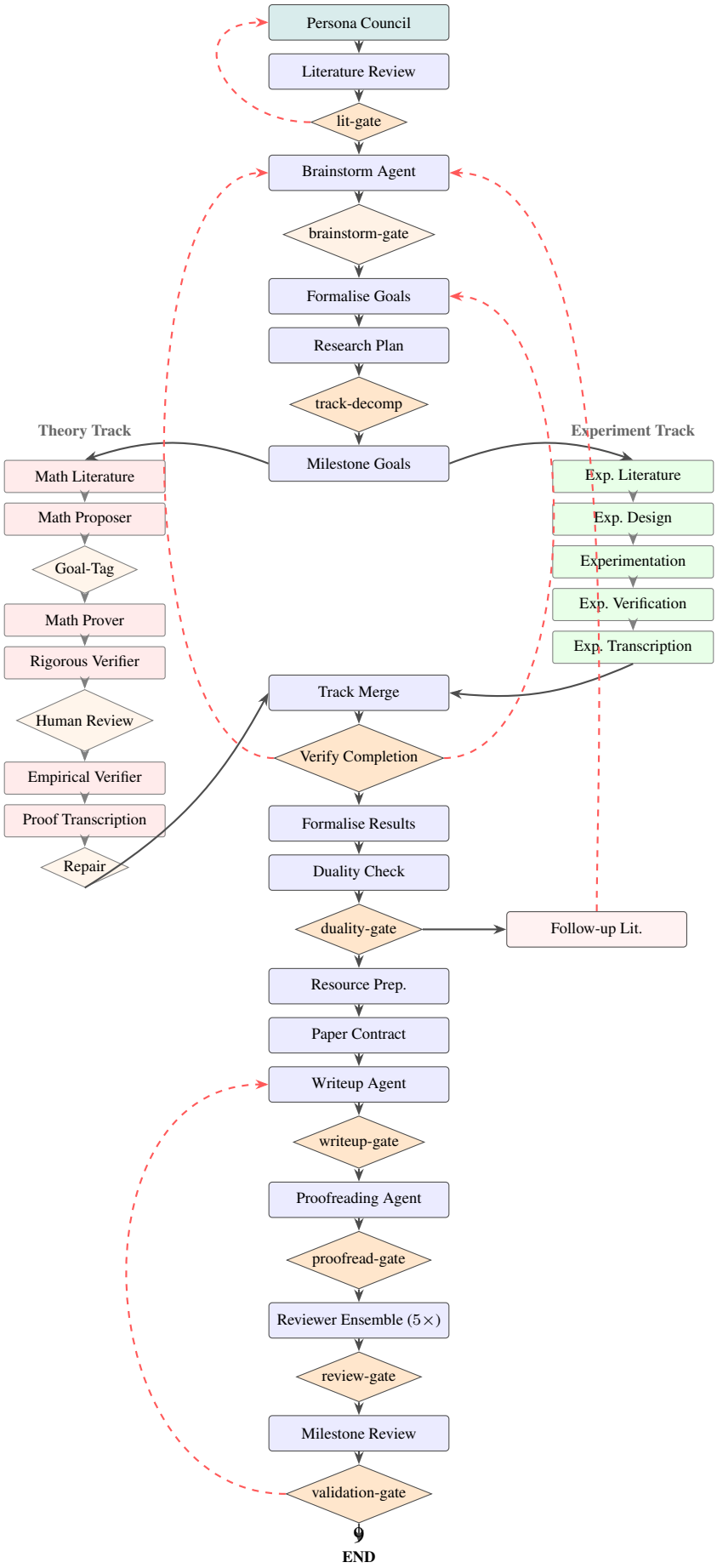

Figure 1: **The pAI agent graph**, audited against the MSc_Internal implementation at commit `6fe0e8e`. Solid arrows are forward edges; dashed red arrows are conditional backtracking routes